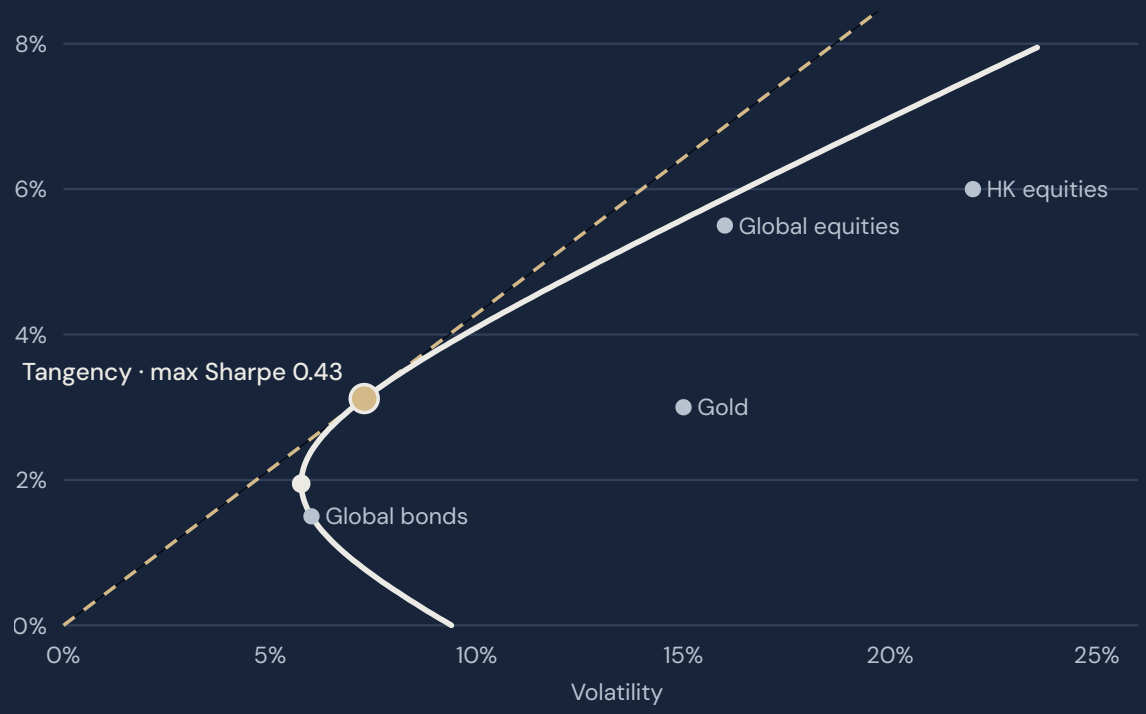
FC FinEx *Careers*



FINEX LEARNING · EXECUTIVE PROGRAMME · QUANTITATIVE FINANCE

Advanced Quantitative Portfolio Management

From factor research to a portfolio you can defend in front of an investment committee.

6 weeks	36	18	23 Nov
online · live + self-paced	CPD hours	lessons · 5 labs	next cohort, 2026

WHY THIS PROGRAMME

From a factor idea to a portfolio you can defend.

Most quantitative courses stop at the model. This programme follows an idea all the way to the trade blotter: where an expected return comes from, how risk is really measured, why the 'optimal' portfolio is so often the worst one out of sample, what a backtest can and cannot prove, and how to explain the result to someone who has to sign off on it.

Each week pairs the core theory, from Markowitz to Black–Litterman to the deflated Sharpe ratio, with a Python lab on real public data. By week six you build, stress-test and present a complete multi-factor portfolio for an Asia-Pacific equity mandate.

Each week includes

- ✓ Short video lessons with worked derivations
- ✓ Interactive models you can manipulate
- ✓ A Python lab on real public data
- ✓ A 90-minute live session, Tuesdays 7:00 pm HKT
- ✓ A knowledge check and curated research

AT A GLANCE

FORMAT	DURATION	EFFORT	CPD HOURS
Online · live + self-paced	6 weeks	≈ 6 hours / week	36
LIVE SESSIONS	**NEXT COHORT**	**TOOLS**	**CERTIFICATE**
Tuesdays 7:00–8:30 pm HKT	Starts 23 November 2026	Python · pandas · cvxpy · scikit-learn	FinEx certificate of completion

Who it is for

- ✓ Portfolio managers and analysts moving from discretionary to systematic processes
- ✓ Risk and performance professionals who want to own the models they review
- ✓ Quantitative researchers who need the portfolio-construction half of the job
- ✓ Wealth and multi-asset teams building model portfolios or factor sleeves

Before you start

- ✓ Comfort with statistics: regression, variance, correlation
- ✓ Working Python, or willingness to complete the free pre-course primer
- ✓ Familiarity with equity markets and basic portfolio concepts

LEARNING OUTCOMES

What you will be able to do.

01

Derive and critique mean–variance, minimum-variance, risk-parity and Black–Litterman portfolios

02

Build, neutralise and combine factor signals into a single expected-return forecast

03

Estimate a covariance matrix that survives out of sample, using shrinkage and factor models

04

Run a backtest that accounts for costs, turnover and multiple testing

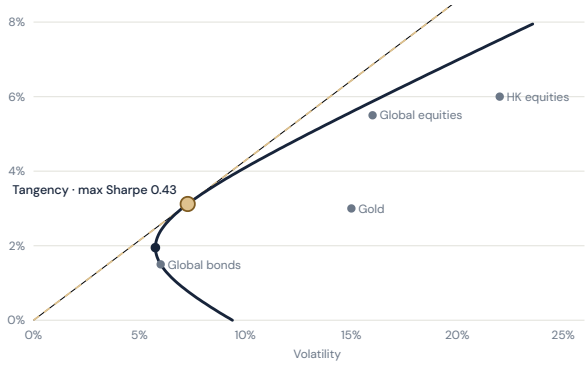
05

Decompose performance into allocation, selection and factor contributions

06

Present a portfolio and its risks to an investment committee with conviction

Week 1 · The efficient frontier and the capital market line



Four illustrative asset classes put through the real mean–variance mathematics. On the course page you can move risk aversion and watch the optimal portfolio slide along the line.

Learn by experimenting

Eight interactive models sit inside the lessons: the efficient frontier, the fundamental law, the research pipeline, covariance shrinkage, risk parity, the deflated Sharpe ratio, market impact and Brinson attribution.

$$w^* = (1/\lambda) \cdot \Sigma^{-1} \mu$$

Every equation is paired with the Python that implements it, using pandas, statsmodels, scikit-learn and cvxpy.

A TYPICAL WEEK

MONDAY

Lessons

2–3 short lessons with derivations, interactive models and code walk-throughs.

TUESDAY

Live session

90 minutes, 7:00 pm HKT: worked problems, Q&A and practitioner case discussion.

WED–SUN

Python lab

Apply the week's method to real public data in a guided notebook.

SUNDAY

Knowledge check

A short check and the week's readings. Completing the week updates your CPD hours.

CURRICULUM · WEEKS 1–3

Foundations, research and construction.

W1

w/c 23 Nov

Portfolio foundations & factor investing

Where expected returns come from, and why diversification is the only free lunch.

1.1 Mean–variance: the idea that started it all · 35 min

1.2 CAPM, factors and the cross-section of returns · 40 min

1.3 The fundamental law of active management · 30 min

Lab Replicate value and momentum

W2

w/c 30 Nov

Building a research workflow in Python

A signal is a pipeline, not a formula. Build one that is honest by construction.

2.1 Data you can trust: point-in-time, survivorship and look-ahead · 35 min

2.2 From raw characteristic to clean signal · 40 min

2.3 Reproducible research · 25 min

Lab Build a composite value-momentum-quality signal

W3

w/c 7 Dec

Risk models and portfolio construction

Why the 'optimal' portfolio is often the worst one, and four ways to fix it.

3.1 The estimation-error problem · 30 min

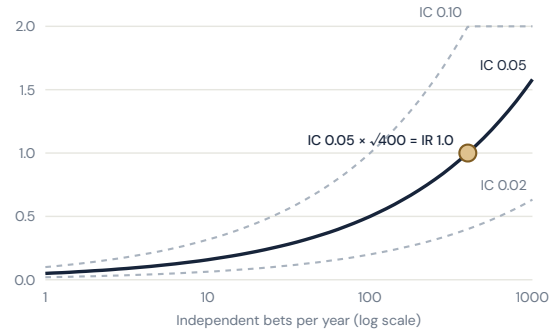
3.2 Shrinkage and factor risk models · 40 min

3.3 Risk budgeting and risk parity · 35 min

3.4 Black–Litterman: blending equilibrium with views · 40 min

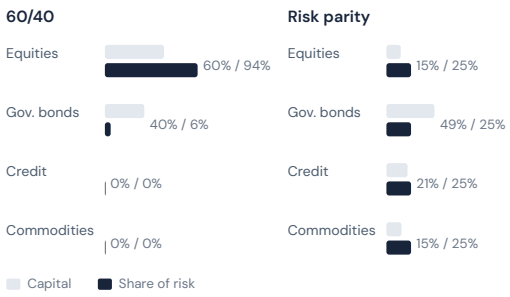
Lab Four portfolios, one universe

Week 1 · The fundamental law



IR ≈ IC · √breadth. A small edge applied many times beats a big edge applied rarely.

Week 3 · Capital vs risk



In a 60/40 portfolio equities carry over 90% of the risk. Risk parity equalises it.

CURRICULUM · WEEKS 4–6

Testing, attribution and the capstone.

W4

w/c 14 Dec

Backtesting and execution strategies

A backtest is not a research tool. It is a way of finding out whether you fooled yourself.

- 4.1 Backtest overfitting and the deflated Sharpe ratio · 40 min
- 4.2 Transaction costs and market impact · 35 min
- 4.3 Optimal execution · 30 min
- 4.4 Trading in Hong Kong: market structure for quants · 35 min
- Lab An honest backtest

W5

w/c 21 Dec

Performance attribution

Was it skill, luck or a factor you did not know you owned?

- 5.1 Brinson attribution · 35 min
- 5.2 Factor-based attribution and risk-adjusted metrics · 35 min
- Lab Explain a decade of returns

W6

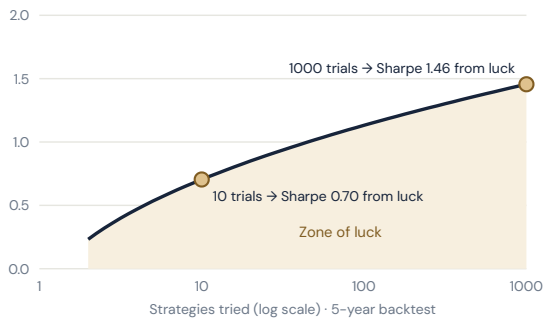
w/c 28 Dec

Final portfolio case study

Put it all together and defend it in front of an investment committee.

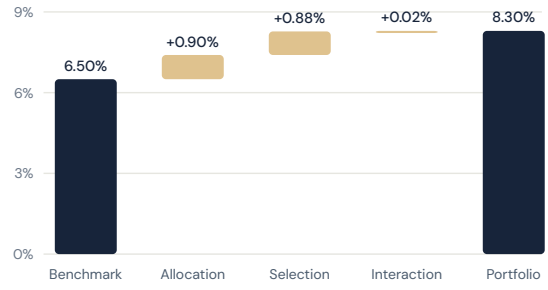
- 6.1 Building the capstone portfolio · 60 min
- 6.2 The investment committee · 45 min
- Lab Capstone presentation

Week 4 · The best Sharpe ratio noise will give you



Bailey & López de Prado (2014). Report the number of trials with every backtest.

Week 5 · Brinson attribution



Worked example: 1.80% active return split into allocation, selection and interaction.

CAPSTONE

An APAC multi-factor equity mandate.

You are handed a HK$2bn Asia-Pacific equity mandate benchmarked to a regional index, with a 4% tracking-error budget and a 30% annual turnover limit. Build a value-momentum-quality portfolio, size it with a risk model you can justify, backtest it honestly, and present it to a mock investment committee.

You will submit

- ✓ A reproducible research notebook (signal → portfolio → backtest)
- ✓ A two-page investment memo with the risk budget and expected tracking error
- ✓ A 10-minute committee presentation with live Q&A

How it is assessed

Research process & reproducibility	**30%**
Portfolio construction & risk	**30%**
Backtest integrity & costs	**20%**
Communication & judgement	**20%**

FACULTY

MV

Dr. Marcus Vance

Former Chief Investment Officer

CFA · Quantitative investing

An illustrative faculty profile focused on translating investment research into practical portfolio decisions, with hands-on modelling and real-world case discussions. Capstones are reviewed by a panel of portfolio managers and risk practitioners.

SCHEDULE

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
w/c 23 Nov	w/c 30 Nov	w/c 7 Dec	w/c 14 Dec	w/c 21 Dec	w/c 28 Dec
Portfolio foundations & factor investing	Building a research workflow in Python	Risk models and portfolio construction	Backtesting and execution strategies	Performance attribution	Final portfolio case study

SIX FORMULAS YOU WILL OWN

MEAN–VARIANCE

$$w^* = (1/\lambda)\,\Sigma^{-1}\,\mu$$

FUNDAMENTAL LAW

$$IR \approx TC \cdot IC \cdot \sqrt{BR}$$

SHRINKAGE

$$\hat{\Sigma} = \delta F + (1 - \delta)S$$

RISK CONTRIBUTION

$$RC_i = w_i(\Sigma w)_i \,/\, \sigma_p$$

MARKET IMPACT

$$\approx Y \cdot \sigma \cdot \sqrt{(Q/V)}$$

BRINSON ALLOCATION

$$(w^p - w^b)(R_i{}^b - R^b)$$

FEES & ENROLMENT

Invest in your next chapter.

PROGRAMME FEE

HK$2,800

FinEx members **HK$2,240** · save 20%

Includes all live sessions, labs and datasets, capstone review and your certificate of completion.

How to enrol

1. Create a free FinEx Careers account
2. Open Learning → Advanced Quantitative Portfolio Management
3. Select **Enrol now** and confirm your details
4. Complete the Python primer before 23 November

Your progress and CPD hours are tracked automatically in **My learning**.

SELECTED READINGS

Markowitz, H. (1952). Portfolio Selection. Journal of Finance, 7(1), 77–91.

Asness, C., Moskowitz, T. & Pedersen, L. (2013). Value and Momentum Everywhere. Journal of Finance, 68(3), 929–985.

Grinold, R. & Kahn, R. (2000). Active Portfolio Management (2nd ed.). McGraw-Hill.

Novy-Marx, R. (2013). The other side of value: The gross profitability premium. Journal of Financial Economics, 108(1), 1–28.

Michaud, R. (1989). The Markowitz Optimization Enigma: Is 'Optimized' Optimal? Financial Analysts Journal, 45(1), 31–42.

DeMiguel, V., Garlappi, L. & Uppal, R. (2009). Optimal Versus Naive Diversification. Review of Financial Studies, 22(5), 1915–1953.

Maillard, S., Roncalli, T. & Teïletche, J. (2010). The Properties of Equally Weighted Risk Contribution Portfolios. Journal of Portfolio Management, 36(4), 60–70.

López de Prado, M. (2018). Advances in Financial Machine Learning. Wiley.

Almgren, R. & Chriss, N. (2000). Optimal execution of portfolio transactions. Journal of Risk, 3(2), 5–39.

Carhart, M. (1997). On Persistence in Mutual Fund Performance. Journal of Finance, 52(1), 57–82.

López de Prado, M. (2016). Building Diversified Portfolios that Outperform Out of Sample. Journal of Portfolio Management, 42(4), 59–69.

Fama, E. & French, K. (1993). Common risk factors in the returns on stocks and bonds. Journal of Financial Economics, 33(1), 3–56.

Harvey, C., Liu, Y. & Zhu, H. (2016). …and the Cross-Section of Expected Returns. Review of Financial Studies, 29(1), 5–68.

Jegadeesh, N. & Titman, S. (1993). Returns to Buying Winners and Selling Losers. Journal of Finance, 48(1), 65–91.

Kenneth R. French Data Library, Tuck School of Business, Dartmouth College (online).

Ledoit, O. & Wolf, M. (2004). Honey, I Shrunk the Sample Covariance Matrix. Journal of Portfolio Management, 30(4), 110–119.

Black, F. & Litterman, R. (1992). Global Portfolio Optimization. Financial Analysts Journal, 48(5), 28–43.

Bailey, D. & López de Prado, M. (2014). The Deflated Sharpe Ratio. Journal of Portfolio Management, 40(5), 94–107.

Perold, A. (1988). The Implementation Shortfall: Paper versus Reality. Journal of Portfolio Management, 14(3), 4–9.

Brinson, G., Hood, L. & Beebower, G. (1986). Determinants of Portfolio Performance. Financial Analysts Journal, 42(4), 39–44.

CFA Institute. Global Investment Performance Standards (GIPS®).

Frazzini, A. & Pedersen, L. (2014). Betting Against Beta. Journal of Financial Economics, 111(1), 1–25.

Illustrative programme. Faculty profile, ratings, dates and fees are sample content. CPD hours are FinEx learning hours and do not imply accreditation by any regulator or professional body. Chart inputs are illustrative; the formulas and readings are real.